Universal Design

Studio Activity

Himanshi Kathuria - 2K24/BD/044
Ilisha Dabas - 2K24/BD/046
Kabir Vijeshwar - 2K24/BD/053

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Activity sheet
Object: Scissors

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Studio Activity Scissors

01. Overview : Standard Scissors

Standard scissors assume two working hands. This excludes people with hemiplegia, limb difference, arthritis, or a temporary injury from a basic task. This case study compares three versions of the same object to show how designing for one group can end up benefiting everyone.

Excessive Force Requirement

Standard household shears require **30 N to 50 N** of grip force to shear **200 GSM** cardstock, leading to joint strain and rapid muscle fatigue for users with limited hand strength.

Complex Multi-Joint Coordination

Operating standard loops requires **3 simultaneous joint** actions (thumb opposition, finger flexion, and wrist stabilization), which completely excludes single-handed or limited-dexterity users.

High Tactile Resistance & Pressure Points

Small plastic loops concentrate force onto a tiny contact area (**under 1.5 cm²**), creating localized pressure points exceeding **200 kPa**, which causes pain during prolonged use.

Lack of Visual & Tactile Contrast

Monochromatic black or metallic handles provide < **10 LRV** (Light Reflectance Value) contrast against standard desktop surfaces, making placement and orientation difficult for low-vision users.

Strict Directional Constraint

Fixed thumb-to-finger orientation forces a strict **90°** wrist angle, preventing activation via forearm, palm, or elbow push.

Core Ergonomic Issues & Engineering Data



02. Assistive Adaptation

Fitted for one user with limited grip strength: a moulded loop handle replaces the finger holes, gripped with a fist instead of two fingers. A return spring drops the reopening effort to **12 N,** so the hand only squeezes, never pulls apart.

It works and it is visibly a repair. Bright therapy-orange plastic over steel reads as medical equipment in a classroom or kitchen, the household pays for it as a single-purpose unit, and it fits only this one grip pattern.

Loop Handle, 34 mm bore
Spring return, 12 N
Squeeze Eases 46 → 28 N

03. Universal Design Version

Ambidextrous loop handle at **30 mm**, adjustable spring tension from **8** to **20 N**, a **30 LRV** contrast strip along the cut line, textured grip zone found by touch. Squeeze force held flat at **18 N** regardless of hand strength, through a longer lever arm.

No adaptation, no separate purchase, no special version. The same detail serves arthritis, a left hand, a child's grip, and someone holding a phone in their other hand.

Loop Handle, 30 mm, ambidextrous
Spring Tension, 8–20 N adjustable
30 LRV cut-line, textured grip

Updated Stage Breakdown with Metrics

A Cut Nobody Has to Adapt For

What the redesign changes:

- Loop handle at **30 mm** - worked by fist, thumb-web, or three fingers together.
- Adjustable spring, **8–20 N** - no manual reopening at any tension setting.
- **30 LRV** contrast plus a textured grip zone, so the cut line is found by low vision or by touch.
- Squeeze force held at **18 N** flat via a longer lever arm - same effort at any grip strength.
- Ambidextrous blade bevel - no dominant-hand penalty.
- The form shows its action - a loop reads as grip, two holes read as nothing.

Mapped to the seven principles:

- **Equitable Use** - One Handle
- **Flexibility -** Fist, Thumb, or Fingers
- **Simple, Intuitive** - Loop Shows Grip
- **Perceptible Info** - Contrast, Texture
- **Tolerance for Error** - Damped Spring Return
- **Low Effort** - 18 N Flat Squeeze
- **Space for Approach** - Ambidextrous Bevel

Criterion	01 Original	02 Adaptation	03 Universal
Usable with one hand	No	**Yes**	**Yes**
Available to every user of the household	**Yes**	No	**Yes**
Needs a separate purchase and fitting	No	**Yes**	No
Marks its user out as different	No	**Yes**	No
Legible without instruction	**Partly**	**Partly**	**Yes**
Works with weak, wet, or gloved grip	No	**Yes**	**Yes**

The adaptation solves one grip's problem and labels it while doing so. The redesign takes the problem out of the object.

A loop handle is not a disability product, it is just a better handle. The test is not whether a person with limited grip can use the scissors, but whether the scissors ever made them ask for something different from what everyone else was given.